

Elasticity

6.1 Elasticity

Elasticity is the property by virtue of which material bodies regain their original shape and size after the external deforming forces are removed. When an external force acts on a body, there is change in its length, shape and volume. The body is said to be strained.

When this external force is removed, the body regains its original shape and size. Such bodies are called elastic bodies. Steel, glass ivory, quartz etc. are elastic bodies. The bodies which do not regain their original shape and size are called plastic bodies. No body is either completely elastic or completely plastic. The property of elasticity is different in different substances. Steel is more elastic than rubber. Liquids and gases are highly elastic.

6.2 Definitions

Elasticity is defined as the property by which a body regains its original position when the forces are withdrawn.

The opposite of elasticity is plasticity. No substance is perfectly elastic or perfectly plastic.

Stress : When a force is applied on a body, there will be relative displacement of the particles and due to the property of elasticity the particles tend to regain their original position. Stress is defined as the restoring force per unit area.

Normal Stress. Restoring force per unit area perpendicular to the surface is called normal stress.

Tangential Stress. Restoring force parallel to the surface per unit area is called tangential stress.

Strain. The ratio of the change in shape to the original shape is called strain. There are three types of strain :

Longitudinal Strain. The ratio of change in length to original length is called longitudinal strain $\left(\frac{l}{L}\right)$

Shearing Strain. Shearing strain is defined as the angle of shear measured in radians.

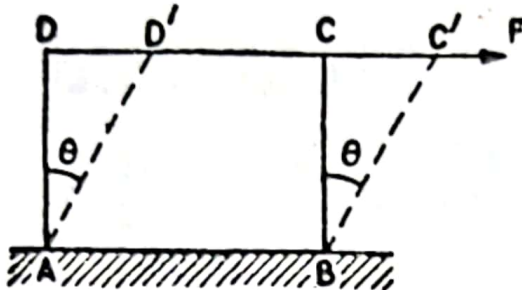


Fig. 6.1

The surface AB is fixed and a force is applied parallel to the surface CD , so that the body is sheared by an angle θ . The angle θ measured in radians is called the shearing strain (θ) (Fig. 6.1).

Volume Strain. The ratio of the change in volume to original

volume is called volume strain $\left(\frac{v}{V}\right)$.

Elastic limit. The maximum stress up to which a body exhibits the property of elasticity is called elastic limit. If the applied force exceeds the maximum stress limit, the body does not regain its original position completely after the external forces are withdrawn.

Hooke's law. It states that within the elastic limit, stress is directly proportional to strain.

$$\text{Stress} \propto \text{Strain}$$

$$\text{Stress} = E \times \text{Strain}$$

$$\therefore \frac{\text{Stress}}{\text{Strain}} = E$$

E is a constant called the modulus of elasticity.

(i) **Young's modulus of elasticity (Y).** It is defined as the ratio of normal stress to longitudinal strain.

$$Y = \frac{\text{Normal Stress}}{\text{Longitudinal Strain}} = \frac{\frac{F}{\text{area}}}{\frac{l}{L}} = \frac{FL}{al}$$

(ii) **Modulus of rigidity (η).** It is defined as the ratio of tangential stress to shearing strain.

$$\eta = \frac{\text{Tangential Stress}}{\text{Shearing Strain}} = \frac{\frac{F}{\text{area}}}{\theta} = \frac{F}{a \times \theta}$$

(iii) **Bulk modulus of elasticity (K).** It is defined as the ratio of normal stress to volume strain,

$$K = \frac{\text{Normal Stress}}{\text{Volume Strain}} = \frac{\frac{F}{\text{area}}}{\frac{v}{V}} = \frac{FV}{a.v} = \frac{pV}{v}$$

where p is the change in pressure.

$$\text{Compressibility} = \frac{1}{K}$$

6.3 Determination of Young's Modulus by Searle's Apparatus

Searle's apparatus consists of a metallic rectangular frame Fig. 6.2. At the positions P and Q two hooks are fixed. One end of a spirit level is fixed with the frame P and the other end rests on the tip of the screw of a spherometer. When the spherometer head is rotated, the free end of the spirit level can be moved up or down. A and B are two long parallel wires, fixed from the same horizontal frame at the top. W is a dead load suspended from the hook of P and keeps the wire B free from kinks. A weight hanger H is attached to the hook of Q .

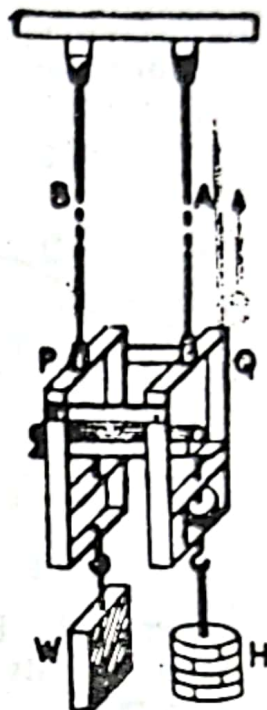


Fig. 6.2

When the wire A is loaded, its length increases and the free end of the spirit level moves down. The head of the spherometer is rotated to bring the spirit level to the horizontal position. The distance by which the spherometer screw is moved gives the extension in the wire for a particular load.

For different loads, the corresponding extensions are obtained by experiment. A graph is drawn between the load (M) along the Y -axis and extension (l) along the X -axis.

Suppose the initial length of the wire A is L cm, the load applied is Mg , radius of the wire is r cm and increase in length is l cm.

$$\text{Then, Stress} = \frac{F}{a} = \frac{Mg}{\pi r^2} \text{ and Strain} = \frac{l}{L}$$

$$\therefore Y = \frac{\text{stress}}{\text{strain}} = \frac{MgL}{\pi r^2 l}$$

The value of $\frac{M}{l}$ is obtained from the graph. The load applied should not be more than the elastic limit of the wire. The elastic limit depends upon the nature of the material and the radius of the wire.

6.4 Glass is more Elastic than Rubber

Glass is more elastic than rubber because for a given stress, the strain produced in glass is much less than that in rubber. Suppose for rubber

$$Y_1 = \frac{FL}{al_1} \quad \dots (i)$$

For glass

$$Y_2 = \frac{FL}{al_2} \quad \dots(ii)$$

For the two materials, F , L and a remaining the same, l_1 for rubber is more than l_2 for glass. Therefore, Y_2 is greater than Y_1 .

6.5 Yield Point, Elastic Limit. Elastic Fatigue

Yield point. When a wire is loaded beyond the elastic limit, Hooke's law is no longer obeyed and the extension produced in the wire is not proportional to the load (Fig. 6.3). The extension is more than the corresponding increase in load. At this stage, the particles of the material go farther apart. Now, if the load is removed, the particles do not regain their original positions. The point B on the graph represents yield point.

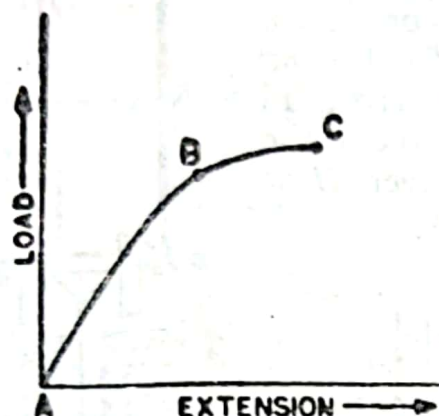


Fig. 6.3

Elastic limit is the maximum stress within which the body exhibits the property of elasticity. Below the elastic limit, the body regains its original position or shape when the deforming force is removed. Beyond the elastic limit, the body does not regain completely its original position even though the external force is withdrawn.

Elastic fatigue. If a body is continuously subjected to stress and strain, it gets fatigued. Consider two torsional pendulums A and B having similar wires. A is initially set into vibration. After A has come to rest, both the pendulums A and B are set into vibration simultaneously. It will be found that due to elastic fatigue, A comes to rest earlier than B .

Example 6.1. Calculate the density of lead under a pressure $2 \times 10^8 \text{ N/m}^2$. Density of lead $= 11.4 \times 10^3 \text{ kg/m}^3$. Bulk modulus of elasticity $= 8 \times 10^9 \text{ N/m}^2$.

$$K = \frac{P}{(-dv/v)}$$

Also

$$M = v\rho$$

Differentiating

$$v d\rho + \rho dv = 0$$

$$\frac{d\rho}{\rho} = - \left(\frac{dv}{v} \right)$$

$\therefore$

$$K = \frac{P}{(d\rho/\rho)}$$

$$d\rho = \frac{p\rho}{K}$$

Here

$$p = 2 \times 10^8 \text{ N/m}^2$$

$$\rho = 11.4 \times 10^3 \text{ kg/m}^3$$

$$K = 8 \times 10^9 \text{ N/m}^2$$

 $\therefore$

$$d\rho = \frac{2 \times 10^8 \times 11.4 \times 10^3}{8 \times 10^9}$$

$$d\rho = 0.285 \times 10^3 \text{ kg/m}^3$$

Density under applied pressure = $\rho + d\rho$

$$= 11.4 \times 10^3 + 0.285 \times 10^3$$

$$= 11.685 \times 10^3 \text{ kg/m}^3$$

Example 6.2. A horizontal steel wire of length 1 m and area of cross section 10^{-6} m^2 is stretched between two fixed supports. When a load of 0.04 kg is applied to the mid point, a sag of $5 \times 10^{-3} \text{ m}$ is produced. For a load of 0.6 kg the sag is $15 \times 10^{-3} \text{ m}$. Calculate the Young's modulus of the wire.

Here

$$\frac{T}{l} = \frac{Mg}{2x}$$

 l is half the length of the wire.

In the first case

$$\frac{T_1}{l_1} = \frac{M_1 g}{2x_1}$$

$$T_1 = \frac{M_1 g l_1}{2x_1}$$

Similarly,

$$T_2 = \frac{M_2 g l_2}{2x_2}$$

$$l_1 = \sqrt{(0.5)^2 + (5 \times 10^{-3})^2}$$

$$l_1 = 0.5(1 + 10^{-4})^{1/2} = 0.5(1 + 0.5 \times 10^{-4})$$

$$l_2 = \sqrt{(0.5)^2 + (15 \times 10^{-3})^2}$$

$$l_2 = 0.5(1 + 10^{-4})^{1/2} = 0.5(1 + 4.5 \times 10^{-4})$$

$$l_2 - l_1 = 0.5(1 + 4.5 \times 10^{-4}) - 0.5(1 + 0.5 \times 10^{-4})$$

$$l_2 - l_1 = 0.5(4 \times 10^{-4})$$

Change in strain

$$= \frac{l_2 - l_1}{l} = \frac{0.5(4 \times 10^{-4})}{0.5} = 4 \times 10^{-4}$$

Change in stress

$$= \frac{T_2 - T_1}{a} = \frac{M_2 g l_2}{2x_2 a} - \frac{M_1 g l_1}{2x_1 a}$$

$$[l_1 = l_2 = 0.5 \text{ m approximately}]$$

$$= \frac{g \times l}{2a} \left(\frac{M_2}{x_2} - \frac{M_1}{x_1} \right)$$

$$\begin{aligned}
 &= \frac{9.8 \times 0.5}{2 \times 10^{-6}} \left(\frac{0.6}{15 \times 10^{-3}} - \frac{0.04}{5 \times 10^{-3}} \right) \\
 &= 2.45 \times 32 \times 10^6 \text{ N/m}^2 \\
 Y &= \frac{\text{Change in stress}}{\text{Change in strain}} = \frac{2.45 \times 32 \times 10^6}{4 \times 10^{-4}} \\
 &= 19.6 \times 10^{10} \text{ N/m}^2
 \end{aligned}$$

6.6 Poisson's Ratio

Whenever a body is subjected to a force in a particular direction, there is change in dimensions of the body in the other two perpendicular directions. This is called lateral strain. Lateral strain is proportional to the size of the body in the other two perpendicular directions.

Let α be the longitudinal strain per unit stress and β the lateral strain per unit stress, Within the elastic limit.

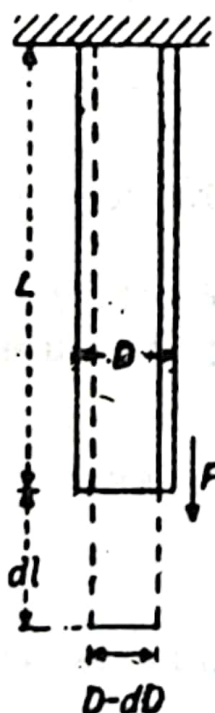
$$\begin{aligned}
 \beta &\propto \alpha \\
 \text{or } \beta &= \sigma \alpha
 \end{aligned}$$

$\therefore$ Poisson's ratio

$$\sigma = \frac{\beta}{\alpha}$$

Therefore, Poisson's ratio is defined as the ratio of lateral strain per unit stress to the longitudinal strain per unit stress.

Consider a wire of length L and diameter D . The wire is fixed at one end and a force is applied at the other end. Consequently the length of the wire increases and the diameter of the wire decreases (Fig. 6.4).



Suppose, increase in length is dL , decrease in diameter is dD .

$$\therefore \sigma = \frac{\beta}{\alpha}$$

$$\sigma = - \frac{dD/D}{dL/L}$$

$$\sigma = - \left(\frac{dD}{dL} \right) \left(\frac{L}{D} \right) \quad \dots(i)$$

The value of Poisson's ratio mainly depends upon the nature of the material of the body. σ has no units as it is a ratio of two numbers. For most of the substances, the value of σ varies between 0.2 and 0.4.

Fig. 6.4 If the volume of the wire remains unchanged after the force has been applied, then

Initial volume of the wire,

$$V = \left(\frac{\pi D^2}{4} \right) L \quad \dots(ii)$$

Differentiating equation (ii)

$$dV = \frac{\pi}{4} [D^2 dL + 2LD dD]$$

If $dV = 0$

$$D^2 dL + 2LD dD = 0$$

$$\frac{dD}{dL} \times \frac{L}{D} = -\frac{1}{2}$$

But $\sigma = -\left(\frac{dD}{dL} \times \frac{L}{D} \right)$

$$\therefore \sigma = -\left(-\frac{1}{2} \right)$$

$$\sigma = \frac{1}{2}$$

This is the maximum possible value of Poisson's ratio.

6.7. Poisson's Ratio of Rubber

A long rubber tube of length about 1 metre is suspended vertically and a weight hanger is fixed at its lower end (Fig. 6.5). The rubber tube is completely filled with water so that the water meniscus is visible in the graduated glass tube C.

When a weight is placed in the weight hanger, the length of the tube increases and its area of cross section decreases. The internal volume of the tube also increases. Consequently the level of water in the tube C falls. The distance through which the pointer P moves is measured with the help of a travelling microscope. Suppose, increase in the length of the tube = dL

Initial volume = V

Initial length = L

Initial area of cross section = A

$$\therefore V = AL$$

After loading

$$(V + dV) = (L + dL)(A + dA)$$

$$dV = (V + dV) - V$$

$$dV = A dL + L dA \quad (i)$$

(neglecting the term $dL \cdot dA$)

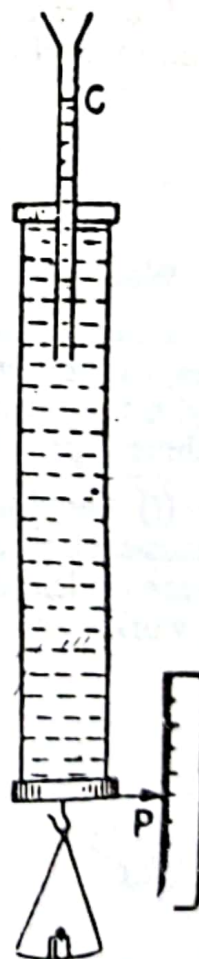


Fig. 6.5

Also

$$A = \frac{\pi D^2}{4}$$

Differentiating

$$dA = \frac{\pi D}{2} dD = \left(\frac{2A}{D}\right) dD$$

Substituting this value of dA in equation (i)

$$dV = AdL - \left(\frac{2AL}{D}\right) dD$$

or

$$\frac{dV}{dL} = A - \left(\frac{2AL}{D}\right) \frac{dD}{dL}$$

$$\therefore \left(\frac{L}{D}\right) \left(\frac{dD}{dL}\right) = \frac{1}{2} \left[1 - \left(\frac{1}{A}\right) \frac{dV}{dL} \right]$$

From definition,

$$\sigma = \frac{\beta}{\alpha} = \frac{dD/D}{dL/L} = \left(\frac{L}{D}\right) \left(\frac{dD}{dL}\right)$$

$$\therefore \sigma = \frac{1}{2} \left[1 - \left(\frac{1}{A}\right) \frac{dV}{dL} \right] \quad \dots(iii)$$

From equation (iii), σ can be calculated.

If a is the area of cross section of the glass tube C and dx is the fall in level of water, then increase in volume

$$dV = a(dx)$$

$$\therefore \sigma = \frac{1}{2} \left[1 - \left(\frac{a}{A}\right) \frac{dx}{dL} \right] \quad \dots(iv)$$

6.8. Work done in Deforming a Body

Whenever a body is deformed by the application of external forces, the body gets strained. The work done is stored in the body in the form of energy and is called the energy of strain. There are three types of strain (i) longitudinal (ii) shearing (iii) volume.

(i) **Longitudinal strain.** Consider a wire of length L , area of cross-section a and Young's modulus of elasticity Y . Let l be the increase in length when a stretching force F is applied. Therefore, work done

$$\int dW = \int_0^l F dl$$

But

$$Y = \frac{F/a}{l/L}$$

or

$$F = \frac{Yal}{L}$$

$$\therefore W = \int_0^l \left(\frac{Ya}{L} \right) l \, dl = \frac{Yal^2}{2L}$$

$$W = \frac{1}{2} \left(\frac{Yal}{L} \right) l = \frac{1}{2} (F) l$$

Work done per unit volume,

$$w = \frac{W}{V} = \frac{W}{aL}$$

$$w = \frac{Fl}{2aL} = \frac{1}{2} \left(\frac{F}{a} \right) \left(\frac{l}{L} \right)$$

$$w = \frac{1}{2} \times \text{Stress} \times \text{Strain}$$

(ii) **Shearing Strain.** Consider a cube of side L . When a tangential force F is applied to the upper face of the cube (keeping the lower face fixed), the cube is sheared through an angle ϕ . If the tangential stress is T ,

$$\eta = \frac{T}{\phi}$$

$$T = \eta\phi$$

Total tangential force,

$$= Ta = \eta a\phi$$

$$\text{Work done,} \quad = \int F \cdot dl,$$

$$\text{Here} \quad dl = L d\phi$$

$$\text{Work done} \quad = \int_0^{\phi} T(L d\phi) = \int_0^{\phi} \eta aL\phi d\phi$$

$$W = \frac{\eta aL\phi^2}{2}$$

Work done per unit volume,

$$w = \frac{\eta aL\phi^2}{2aL} = \frac{\eta\phi^2}{2} = \frac{(\eta\phi)\phi}{2}$$

$$w = \frac{1}{2} (T) \phi$$

$$w = \frac{1}{2} \times \text{Stress} \times \text{Strain}$$

(iii) **Volume Strain.** Consider a cube of volume V , area of cross-section A and length L . When a normal stress p is applied, the change in volume $= v$

$$\text{Work done} = \int_0^v p \cdot dv$$

$$\text{But } K = \frac{pV}{v}$$

$$\text{or } p = \frac{Kv}{V}$$

$$\therefore W = \int_0^v \frac{Kv}{V} dv$$

$$W = \frac{Kv^2}{2V} = \frac{1}{2} \left(\frac{Kv}{V} \right) v$$

$$W = \frac{1}{2} p \times v$$

Work done per unit volume,

$$w = \frac{p \times v}{2V} = \frac{1}{2} (p) \left(\frac{v}{V} \right)$$

$$w = \frac{1}{2} \times \text{Stress} \times \text{Strain} \quad \dots(iii)$$

Example 6.3 The young's modulus of a metal is $2 \times 10^{11} \text{ N/m}^2$ and its breaking stress is $1.078 \times 10^9 \text{ N/m}^2$. Calculate the maximum amount of energy per unit volume which can be stored in the metal when stretched. [Calcutta, 1980]

$$\text{Here } Y = 2 \times 10^{11} \text{ N/m}^2$$

$$\text{Maximum Stress} = 1.078 \times 10^9 \text{ N/m}^2$$

$$\text{Energy stored per unit volume} = \frac{1}{2} \times \text{stress} \times \text{strain}$$

$$= \left[\frac{1}{2} \times 1.078 \times 10^9 \right] \left[\frac{\text{Stress}}{Y} \right]$$

$$= \frac{1.078 \times 10^9 \times 1.078 \times 10^9}{2 \times 2 \times 10^{11}} = 2.90 \times 10^6 \text{ J/m}^3$$

Example 6.4. Find the work done in stretching a wire of 1 sq mm cross section and 2 m long through 0.1 mm. $Y = 2 \times 10^{11} \text{ N/m}^2$. (Delhi, 1976)

Here

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

$$L = 2 \text{ m}$$

$$l = 0.1 \text{ mm} = 10^{-4} \text{ m}$$

$$a = 1 \text{ sq mm} = 10^{-6} \text{ m}^2$$

Strain

$$= \frac{l}{L} = \frac{10^{-4}}{2} = 5 \times 10^{-5}$$

Stress

$$= Y \times \text{Strain}$$

Stress

$$= 2 \times 10^{11} \times 5 \times 10^{-5}$$

Stress

$$= 10^7 \text{ N/m}^2$$

Volume of the wire

$$= aL = 10^{-6} \times 2 = 2 \times 10^{-6} \text{ m}^3$$

Work done per unit volume = $\frac{1}{2} \times \text{Stress} \times \text{Strain}$

Total work done = $\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{Volume}$

$$\therefore W = \frac{1}{2} \times 10^7 \times 5 \times 10^{-5} \times 2 \times 10^{-6} \text{ Joules}$$

$$W = 5 \times 10^{-4} \text{ J}$$

Example 6.5. Calculate the work done in stretching a uniform metal wire of area of cross section 10^{-6} m^2 and length 1.5 m through $4 \times 10^{-3} \text{ m}$. Given, $Y = 2 \times 10^{11} \text{ N/m}^2$. (Calcutta, 1976)

Here

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

$$L = 1.5 \text{ m}$$

$$a = 10^{-6} \text{ m}^2$$

$$l = 4 \times 10^{-3} \text{ m}$$

Strain

$$\frac{l}{L} = \frac{4 \times 10^{-3}}{1.5}$$

Stress

$$= Y \times \text{Strain}$$

Stress

$$= \frac{2 \times 10^{11} \times 4 \times 10^{-3}}{1.5} \text{ N/m}^2$$

Volume of the wire = $aL = 10^{-6} \times 1.5 \text{ m}^3$

Work done per unit volume = $\frac{1}{2} \times \text{Stress} \times \text{Strain}$

Total work done = $\frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$

$$W = \frac{1}{2} \times \left[\frac{2 \times 10^{11} \times 4 \times 10^{-3}}{1.5} \right] \times \left(\frac{4 \times 10^{-3}}{1.5} \right) \times 1.5 \times 10^{-6}$$

$$W = 1.066 \text{ J}$$

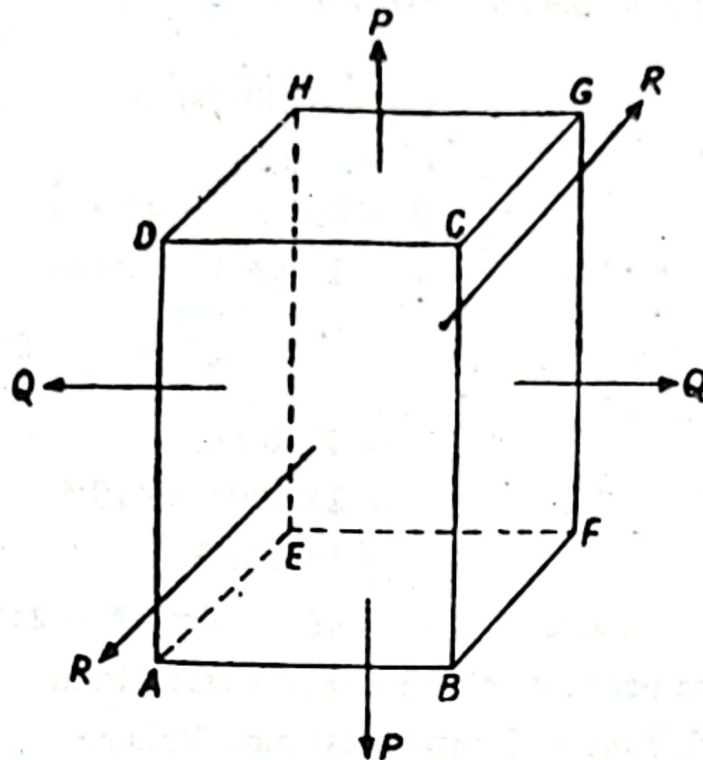
6.9. Bulk Modulus. (Relation between K , γ and σ)

Fig. 6.6.

Consider a block of rectangular cross section having length l , breadth b and thickness t . Let the block be subjected to outward stress as shown in Fig. 6.6. Here, P is the stress acting on the faces $ABFE$ and $CDHG$. Here Q is the stress acting on the faces $AEHD$ and $BCGF$ and R is the stress acting on the faces $ABCD$ and $EFGH$.

Each stress produces an extension in its own direction and a lateral contraction in the other two perpendicular directions. Let α be the longitudinal strain per unit stress and β the lateral strain per unit stress. Poisson's ratio $\sigma = \frac{\beta}{\alpha}$

Increase in length due to the stress P is Pal and corresponding contraction in breadth and thickness will be $P\beta b$ and $P\beta t$ respectively.

Similarly, due to the stress Q , increase in breadth is Qab and corresponding decrease in length and thickness will be $Q\beta l$ and $Q\beta t$ respectively. Also due to the stress R , increase in thickness is Rat and decrease in length and breadth will be $R\beta l$ and $R\beta b$.

Final length	$= l + Pal - Q\beta l - R\beta l$
Final breadth	$= b + Qab - P\beta b - R\beta b$
Final thickness	$= t + Rat - P\beta t - Q\beta t$
Final volume	$= lbt[1 + Pa - (Q + R)\beta]$
	$\times [1 + Qa - (P + R)\beta] \times [1 + Ra - (P + Q)\beta]$
	$= lbt[1 + (P + Q + R)\alpha - 2\beta(P + Q + R)]$
	$= lbt[1 + (P + Q + R)(\alpha - 2\beta)]$

$$\text{Change in volume} = lbt(P+Q+R)(\alpha-2\beta)$$

If the stresses are equal, $P = Q = R$

$$\text{Change in volume} = lbt(3P)(\alpha-2\beta)$$

$$\text{Strain} = \frac{lbt(3P)(\alpha-2\beta)}{lbt} = 3P(\alpha-2\beta)$$

$$\text{Bulk modulus, } K = \frac{P}{3P(\alpha-2\beta)}$$

$$\text{or } K = \frac{1}{3(\alpha-2\beta)} \quad \dots(i)$$

$$\text{or } K = \frac{1}{3\alpha\left(1-\frac{2\beta}{\alpha}\right)}$$

$$K = \frac{1}{3\alpha(1-2\sigma)} \quad \dots(ii)$$

$$\text{But } \frac{1}{\alpha} = Y$$

$$K = \frac{Y}{3(1-2\sigma)} \quad \dots(iii)$$

6.10 Modulus of Rigidity

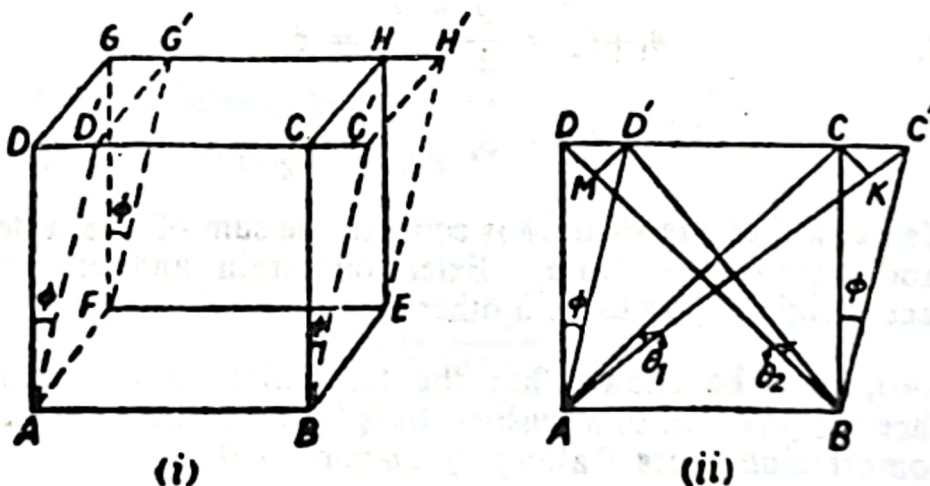


Fig. 6.7

Consider a cubical block $ABCDEFGH$ of length l . The face $ABEF$ is kept fixed and a tangential stress T is applied on the face $CDGH$. The block gets sheared through an angle ϕ [Fig. 6.7 (i)]. All the particles in the plane $CDGH$ remain in the same plane and displaced equally in the same direction. Consequently, D is shifted to D' , C to C' , G to G' , and H to H' .

The diagonal AC gets extended to AC' while the diagonal BD gets compressed to BD' . Extension strain is θ and also compression strain is θ . [Fig. 6.7 (ii)]

As the angle of shear ϕ is very small, the distance between $ABEF$ and $C'D'G'H'$ practically remains the same. Therefore, for a small shear angle ϕ , $\angle CC'B = 90^\circ$ and $CC'K = 45^\circ$.

$$CK = C'K = CC' \cos 45^\circ = \frac{CC'}{\sqrt{2}}$$

$$\text{Similarly } DM = D'M = \frac{DD'}{\sqrt{2}}$$

$$\text{If } CC' = DD' = l$$

$$CK = C'K = \frac{l}{\sqrt{2}}$$

$$\text{and } DM = D'M = \frac{l}{\sqrt{2}}$$

$$\text{Extension strain} = \theta_1 = \frac{CK}{AC} = \frac{\frac{l}{\sqrt{2}}}{\frac{l}{2L}} = \frac{l}{2L} = \frac{\phi}{2}$$

$$\text{Compression strain} = \theta_2 = \frac{D'M}{B'D} = \frac{\frac{l}{\sqrt{2}}}{\frac{l}{2L}} = \frac{l}{2L} = \frac{\phi}{2}$$

$$\therefore \theta_1 = \theta_2 = \theta$$

$$\therefore \theta_1 + \theta_2 = \frac{\phi}{2} + \frac{\phi}{2} = \phi$$

$$\text{or } 2\theta = \phi, \quad \theta = \frac{\phi}{2}$$

Hence, a shearing strain ϕ is equal to the sum of the extension strain and compression strain. Extension strain and compression strain act at right angles to each other.

Also, it can be shown that the tangential stress T along the upper face is equivalent to a tensile stress T along the diagonal AC and a compression stress T along the diagonal BD .

Relation between η , Y , and σ

Let α and β be the longitudinal and lateral strains per unit stress respectively.

Extension along the diagonal AC due to the tensile stress

$$= T\alpha(AC)$$

There is also lateral extension along AC due to the compression stress T along the diagonal BD and it is equal to $T\beta(AC)$

$$\text{Total extension along } AC = T(AC)(\alpha + \beta)$$

Elasticity

$$CK = T(AC) (\alpha + \beta)$$

$$\frac{l}{\sqrt{2}} = (L\sqrt{2}) (\alpha + \beta)$$

$$\frac{l}{L} = 2T (\alpha + \beta)$$

But

$$\phi = \frac{l}{L}$$

$\therefore$

$$\phi = 2T(\alpha + \beta)$$

or

$$\frac{T}{\phi} = \frac{1}{2(\alpha + \beta)}$$

$\therefore$

$$\eta = \frac{T}{\phi} = \frac{1}{2(\alpha + \beta)}$$

Hence

$$\eta = \frac{1}{2(\alpha + \beta)} \quad \dots (i)$$

$$\eta = \frac{1}{2\alpha(1 + \sigma)}$$

But

$$\frac{1}{\alpha} = Y$$

$\therefore$

$$\eta = \frac{Y}{2(1 + \sigma)} \quad \dots (ii)$$

6.11 Relation between Elastic Constants (Y , η , K and σ)

The various equations are

$$Y = \frac{1}{\alpha} \quad \dots (i)$$

$$K = \frac{1}{3(\alpha - 2\beta)} \quad \dots (ii)$$

$$\eta = \frac{1}{2(\alpha + \beta)} \quad \dots (iii)$$

From equations (ii) and (iii)

$$\alpha - 2\beta = \frac{1}{3K} \quad \dots (iv)$$

and

$$2\alpha + 2\beta = \frac{1}{\eta} \quad \dots (v)$$

Adding (iv) and (v) and simplifying

$$3\alpha = \frac{1}{3K} + \frac{1}{\eta}$$

$$\frac{3}{Y} = \frac{1}{3K} + \frac{1}{\eta} \quad \dots (vi)$$

From equation (vi)

$$Y = \frac{9\eta K}{\eta + 3K} \quad \dots(vii)$$

$$\eta = \frac{3KY}{9K - Y} \quad \dots(viii)$$

$$K = \frac{Y\eta}{9\eta - 3Y} \quad \dots(ix)$$

From equations (vii), (viii) and (ix), values of Y , η and K can be calculated if any two values are known.

From equation (iii)

$$\eta = \frac{1}{2(\alpha + \beta)} = \frac{1}{2\alpha \left(1 + \frac{\beta}{\alpha}\right)}$$

$$\eta = \frac{Y}{2(1 + \sigma)} \quad \dots(x)$$

or $\sigma = \left(\frac{Y}{2\eta}\right) - 1 \quad \dots(xi)$

Multiplying equation (ix) by 2 and subtracting it from equation (v) we get

$$6\beta = \frac{1}{\eta} - \frac{2}{3K}$$

or $\beta = \frac{3K - 2\eta}{18\eta K} \quad \dots(xii)$

The relation for Poisson's ratio can be found in terms of K and η .

$$\sigma = \frac{\beta}{\alpha} = \beta Y = \left(\frac{3K - 2\eta}{18\eta K}\right) \left(\frac{9\eta K}{\eta + 3K}\right)$$

$$\sigma = \frac{3K - 2\eta}{2(\eta + 3K)} \quad \dots(xiii)$$

6.12 Alternative Method

The relation between Y , η and K can also be obtained with the help of the following table.

Consider a unit cube, which is subjected to outward elongational force P on each face. Let σ be the Poisson's ratio for the material. In the table, the values of applied stress and the corresponding strains produced along the three perpendicular axes are shown. For a stress P the longitudinal strain produced $= \frac{P}{Y}$ in its own direction and the corresponding strains in the other two

perpendicular directions are $-\frac{\sigma P}{Y}$ and $-\frac{\sigma P}{Y}$. Negative sign indicates contraction.

Stress along			Strain along				
OX	OY	OZ	OX	OY	OZ		
$+P$	0	0	$+\frac{P}{Y}$	$-\frac{\sigma P}{Y}$	$-\frac{\sigma P}{Y}$	Total volume strain	
0	$+P$	0	$-\frac{\sigma P}{Y}$	$\frac{P}{Y}$	$-\frac{\sigma P}{Y}$		
0	0	$+P$	$-\frac{\sigma P}{Y}$	$-\frac{\sigma P}{Y}$	$\frac{P}{Y}$		
Volume	$+P$	$+P$	$+P$	$\frac{P}{Y} (1-2\sigma)$	$\frac{P}{Y} (1-2\sigma)$	$\frac{P}{Y} (1-2\sigma)$	$=\frac{3P}{Y} (1-2\sigma)$
Shearing	$+P$	0	0	$+\frac{P}{Y}$	$-\frac{\sigma P}{Y}$	$-\frac{\sigma P}{Y}$	Shearing Strain
	0	$-P$	0	$+\frac{\sigma P}{Y}$	$-\frac{P}{Y}$	$+\frac{\sigma P}{Y}$	
Sum	$+P$	$-P$	0	$\frac{P}{Y} (1+\sigma)$	$-\frac{P}{Y} (1+\sigma)$	0	$=\frac{2P}{Y} (1+\sigma)$

For an isotropic material,

$$\text{Volume strain} = \frac{P}{K}$$

From the table,

$$\text{Volume strain} = \frac{3P}{Y} (1-2\sigma)$$

$$\therefore \frac{P}{K} = \frac{3P}{Y} (1-2\sigma)$$

$$K = \frac{Y}{3(1-2\sigma)} \quad \dots(i)$$

$$\text{Shearing strain} = \frac{P}{\eta}$$

From the table,

$$\text{Shearing strain} = \frac{2P}{Y} (1+\sigma)$$

$$\therefore \frac{P}{\eta} = \frac{2P(1+\sigma)}{Y}$$

$$\text{or} \quad \eta = \frac{Y}{2(1+\sigma)} \quad \dots(ii)$$

From equations (i) and (ii) we get

$$1-2\sigma = \frac{Y}{3K} \quad \dots(iii)$$

$$\text{and} \quad 2+2\sigma = \frac{Y}{\eta} \quad \dots(iv)$$

Adding (iii) and (iv)

$$3 = Y \left[\frac{1}{3K} + \frac{1}{\eta} \right]$$

$$\text{or} \quad \frac{3}{Y} = \frac{1}{3K} + \frac{1}{\eta} \quad \dots(v)$$

6.13 Limiting Values of σ

From Article 6.11,

$$\eta = \frac{1}{2(\alpha+\beta)} = \frac{1}{2\alpha \left(1 + \frac{\beta}{\alpha} \right)}$$

$$\eta = \frac{Y}{2(1+\sigma)} \quad \dots(i)$$

$$\text{and} \quad K = \frac{1}{3(\alpha-2\beta)} = \frac{1}{3\alpha \left(1 - \frac{2\beta}{\alpha} \right)}$$

$$\text{or} \quad K = \frac{Y}{3(1-2\sigma)} \quad \dots(ii)$$

From equations (i) and (ii)

$$Y = 3K(1-2\sigma) = 2\eta(1+\sigma)$$

$$i. e. \quad 3K(1-2\sigma) = 2\eta(1+\sigma) \quad \dots(iii)$$

(a) If the Poisson's ratio is a positive quantity, as K and η are always positive,

$$(1-2\sigma) > 0$$

$$2\sigma < 1 \text{ or } \sigma < 0.5$$

(b) If the Poisson's ratio is a negative quantity, for η and K to be positive,

$$(1+\sigma) > 0$$

$$\sigma > -1$$

It means the value σ lies between -1 and $+0.5$

However, in actual practice the value of σ cannot be negative because the body does not expand laterally when it expands longitudinally. Also when $\sigma = 0.5$, it means that there is no change in volume and the body is completely incompressible. Therefore, it is not possible. In practice, the value of σ for most of the isotropic substances is between 0.2 and 0.4 .

Example 6.6 The modulus of rigidity and Poisson's ratio of the material of a wire are $2.87 \times 10^{10} \text{ N/m}^2$ and 0.379 respectively. Find the value of young's modulus of the material of the wire.

[Calcutta, 1978]

Hence

$$\eta = 2.87 \times 10^{10} \text{ N/m}^2$$

$$\sigma = 0.379$$

$$\eta = \frac{Y}{2(1+\sigma)}$$

$$Y = 2\eta(1+\sigma)$$

$$= 2 \times 2.87 \times 10^{10} (1+0.379)$$

$$= 7.915 \times 10^{10} \text{ N/m}^2$$

Example 6.7 A steel wire 8 m long and 4 mm in diameter is fixed to two rigid supports. Calculate the increase in tension when the temperature falls by 10°C .

$$\alpha = 12 \times 10^{-6}/^\circ\text{C}$$

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

Here

$$\alpha = 12 \times 10^{-6}/^\circ\text{C}$$

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

$$t = 10^\circ\text{C}$$

$$L = 8 \text{ m}$$

$$r = 2 \times 10^{-3} \text{ m}$$

$$\text{Increase in length, } l = L \alpha t$$

$$\text{Strain} = \frac{l}{L} = \frac{L \alpha t}{L} = \alpha t$$

$$\text{Stress} = Y \times \text{Strain}$$

$$= Y \alpha t$$

$$\text{Area of cross-section} = \pi r^2$$

$$\text{Increase in tension} = \text{stress} \times \text{area of cross-section}$$

$$= Y \alpha t \times \pi r^2 \text{ N}$$

$$= \frac{2 \times 10^{11} \times 12 \times 10^{-6} \times 10 \times 22 \times (2 \times 10^{-3})^2}{7}$$

$$= 301.7 \text{ N}$$

Example 6.8. A steel rod of length 5 m is fixed rigidly between two supports. The coefficient of linear expansion of steel $= 12 \times 10^{-6}/^\circ\text{C}$. Calculate the stress in the rod for an increase in temperature of 40°C . The Young's modulus of elasticity of steel $2 \times 10^{11} \text{ N/m}^2$.

Here

$$\alpha = 12 \times 10^{-6}/^\circ\text{C},$$

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

$$t = 40^\circ\text{C}$$

$$L = 5 \text{ m}$$

Increase in length, $l = L \alpha t$

Strain

$$= \frac{l}{L} = \frac{L \alpha t}{L}$$

$$= \alpha t$$

$$\text{Stress} = Y \times \text{Strain} = Y \alpha t$$

$$= 2 \times 10^{11} \times 12 \times 10^{-6} \times 40$$

$$= 9.6 \times 10^7 \text{ N/m}^2$$

Example 6.9 A uniform rod of 2 sq mm cross-section is heated from 0°C . to 20°C . Find the force which must be exerted to prevent it from expanding. Find also the energy stored per unit volume. $\alpha = 12 \times 10^{-6}/^\circ\text{C}$ and $Y = 10^{11}$ newtons/sq metre.

Here

$$\alpha = 12 \times 10^{-6}/^\circ\text{C},$$

$$Y = 10^{11} \text{ N/m}^2$$

$$t = 20^\circ\text{C},$$

$$a = 2 \text{ sq mm} = 2 \times 10^{-6} \text{ sq. metre}$$

Increase in length, $l = L \alpha t$

$$\text{Strain} = \frac{l}{L} = \frac{L \alpha t}{L} = \alpha t$$

$$\text{Stress} = Y \times \text{Strain} = Y \alpha t$$

$$\text{Force} = \text{Stress} \times \text{area of cross-section}$$

$$= Y \alpha t \times a$$

$$= 10^{11} \times 12 \times 10^{-6} \times 20 \times 2 \times 10^{-6}$$

$$= 48 \text{ newtons}$$

Energy stored per unit volume

$$= \frac{1}{2} \text{ stress} \times \text{strain} = \frac{1}{2} \times Y \alpha t \times \alpha t$$

$$= \frac{10^{11} \times (12 \times 10^{-6})^2 \times (20)^2}{2}$$

$$= 2880 \text{ J/m}^3.$$

Example 6.10. A steel wire of diameter 1 mm supports a load which is sufficient to keep the wire taut at 20°C . Calculate the additional load that will be required to restore the length of the wire to its initial value when the temperature falls to 0°C .

$$Y = 2 \times 10^{11} \text{ newtons/sq metre}$$

$$\alpha = 10^{-6}/^\circ\text{C},$$

$$\text{diameter} = 1 \text{ mm} = 10^{-3} \text{ metre}$$

Area of cross-section

$$a = \frac{\pi d^2}{4} = \frac{22 \times 10^{-6}}{28} \text{ sq metre}$$

$$\text{Decrease in length} = L \alpha t$$

$$\text{Strain} = \frac{L \alpha t}{L} = \alpha t$$

$$\text{Stress} = Y \times \text{Strain} = Y \alpha t$$

Force required to maintain the original length

$$= \text{Stress} \times \text{area of cross-section}$$

$$= Y \alpha t \times a$$

$$= \frac{2 \times 10^{11} \times 12 \times 10^{-6} \times 20 \times 22 \times 10^{-6}}{28} \text{ newtons}$$

$$= \frac{44 \times 6}{7 \times 9.8} \text{ kg wt}$$

$$= 3.847 \text{ kg wt}$$

Example 6.11. Calculate the longest length of a steel wire that can hang vertically without breaking. Breaking stress for steel $= 8 \times 10^8 \text{ N/m}^2$ and density of steel $= 8 \times 10^3 \text{ kg/m}^3$.

Let l be the longest length of the wire

$$\text{Area of cross section} = a$$

$$\text{Mass of the wire} = L \alpha \rho$$

$$\text{Weight of the wire} = (L \alpha \rho) g$$

$$\text{Stress on the wire} = \frac{(L a \rho) g}{a} = L \rho g$$

$$L \rho g = 8 \times 10^3$$

$$L = \frac{8 \times 10^3}{\rho g}$$

But $\rho = 8 \times 10^3 \text{ kg/m}^3$

$$L = \frac{8 \times 10^3}{8 \times 10^3 \times 9.8} = 1.021 \times 10^{-4} \text{ m}$$

Example 6.12. Calculate the Poisson's ratio for the material, given $Y = 12.25 \times 10^{10} \text{ N/m}^2$ and $\eta = 4.55 \times 10^{10} \text{ N/m}^2$.
(Gorakhpur, 1971)

Here

$$\eta = \frac{1}{2(\sigma + \beta)} = \frac{1}{2\alpha \left(1 + \frac{\beta}{\alpha}\right)}$$

But $Y = \frac{1}{\alpha}$

and $\sigma = \frac{\beta}{\alpha}$

$$\therefore \eta = \frac{Y}{2(1 + \sigma)}$$

$$\therefore \sigma = \left(\frac{Y}{2\eta}\right) - 1$$

Here, $Y = 12.25 \times 10^{10} \text{ N/m}^2$

$$\eta = 4.55 \times 10^{10} \text{ N/m}^2$$

$$\therefore \sigma = \left(\frac{12.25 \times 10^{10}}{2 \times 4.55 \times 10^{10}}\right) - 1$$

$$\sigma = 1.347 - 1$$

$$\sigma = 0.347$$

6.14 Twisting of a Cylinder

Consider a cylindrical rod of length l and radius r . Its upper end is clamped and the rod is twisted by applying a couple to its lower end in a plane perpendicular to its length. The rod is said to be under torsion. A reaction is set up due to its property of elasticity and a restoring couple equal and opposite to the twisting couple is produced (Fig. 6.8).

All the particles in the lower end are shifted through the same angle θ but the linear displacement of a particle near the rim is more

than the particle near the centre. Hence shearing angle ϕ is more for the particles near the rim than near the axis of the cylinder.

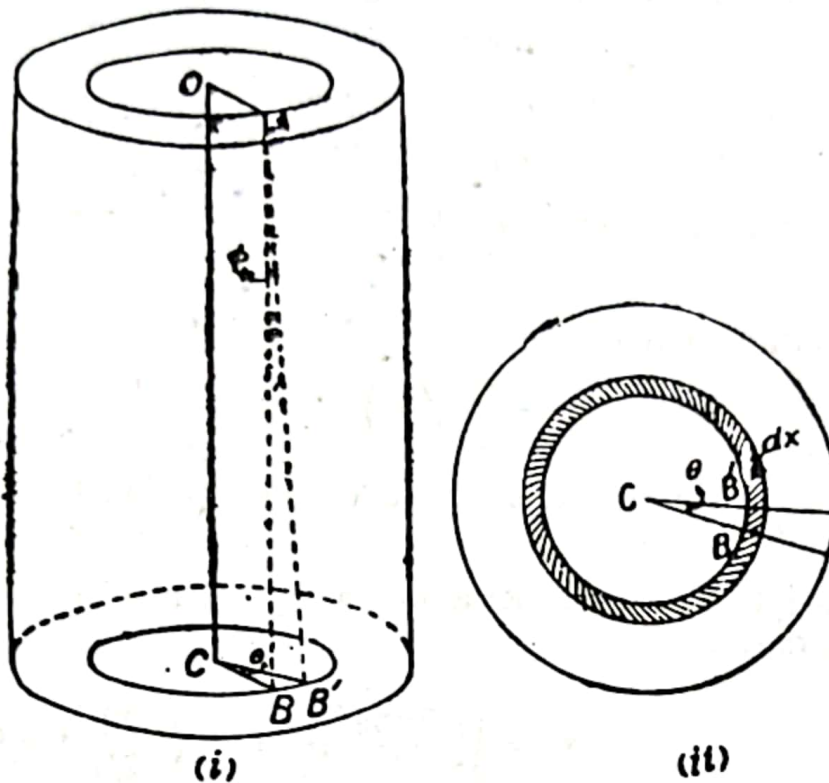


Fig. 6.8

Consider a cylindrical shell of radius x and radial thickness dx [Fig. 6.8 (ii)].

$$BB' = x\theta = l\phi$$

$$\text{or } \phi = \left(\frac{x\theta}{l} \right) \quad \dots(i)$$

ϕ is maximum at the rim and zero at the axis.

$$\text{Shearing stress} = T$$

$$\eta = \frac{T}{\phi}$$

$$\therefore T = \eta\phi = \left(\frac{\eta x\theta}{l} \right)$$

$$\text{Area of cross section of the shell} = 2\pi x dx$$

$$\text{Shearing force on the shell} = T \times \text{area}$$

$$F = \left(\frac{\eta x\theta}{l} \right) 2\pi x dx$$

$$F = \left(\frac{2\pi\eta\theta}{l} \right) x^2 dx$$

Moment of this force about the axis OC,

$$= \left[\left(\frac{2\pi\eta\theta}{l} \right) x^2 dx \right] x$$

$$= \left(\frac{2\pi\eta\theta}{l} \right) x^3 dx$$

Total twisting couple for the whole cylinder

$$\tau = \int_0^r \left(\frac{2\pi\eta\theta}{l} \right) x^3 dx = \left(\frac{\pi\eta r^4}{2l} \right) \theta \quad \dots(ii)$$

Let the couple per unit twist be C

$$\tau = C\theta$$

Comparing (i) and (ii)

$$C = \left(\frac{\pi\eta r^4}{2l} \right) \quad \dots(iii)$$

Couple per unit twist C is also called the torsional rigidity of the material of the wire.

Hollow cylinder. Consider a hollow cylinder of inner radius r_2 and outer radius r_1 .

$$\therefore \tau' = \int_{r_2}^{r_1} \left(\frac{2\pi\eta\theta}{l} \right) x^3 dx = \left(\frac{\pi\eta\theta}{2l} \right) (r_1^4 - r_2^4)$$

$$\therefore C' = \frac{\pi\eta}{2l} (r_1^4 - r_2^4) \quad \dots(iv)$$

Special Case. If two cylinders of the same length and same mass and of the same material are made such that one cylinder is solid of radius r and the other cylinder is hollow of inner radius r_2 and outer radius r_1 then,

$$\pi r^2 = \pi(r_1^2 - r_2^2)$$

In the case of a solid cylinder,

$$C = \frac{\pi\eta r^4}{2l} = \frac{\pi\eta(r_1^2 - r_2^2)^2}{2l} \quad \dots(i)$$

For a hollow cylinder

$$C' = \frac{\pi\eta(r_1^4 - r_2^4)}{2l} = \frac{\pi\eta(r_1^2 + r_2^2)(r_1^2 - r_2^2)}{2l} \quad \dots(ii)$$

Dividing (ii) by (i)

$$\frac{C'}{C} = \frac{r_1^2 + r_2^2}{r_1^2 - r_2^2} \quad \dots(iii)$$

$$\therefore C' > C$$

Hence, a larger twisting couple will be required in the case of a hollow cylinder as compared to a solid cylinder of the same material, mass and length.

$$\eta = \frac{4\pi MR^2 l}{r^2 t^2}$$

Here

$$M = 1.2 \text{ kg}$$

$$R = 0.05 \text{ m}$$

$$l = 1 \text{ m}$$

$$r = 0.72 \times 10^{-3} \text{ m}$$

$$t = 1.98 \text{ s}$$

$\therefore$

$$\eta = \frac{4\pi \times 1.2 \times (0.05)^2 \times 1}{(0.72 \times 10^{-3})^4 \times (1.98)^2}$$

$$\eta = 3.579 \times 10^{10} \text{ N/m}^2.$$

6.18 Bending of Beams

A beam is defined as a structure of uniform cross section, whose length is large in comparison to its breadth and thickness. For such a structure, the shearing stress for any given cross section is negligible. Beams are used in the construction of bridges or for purposes of supporting heavy loads. They are commonly used in the structure of multistoried buildings. A beam can be used in a horizontal position or as columns and pillars in a vertical position.

Consider a beam of uniform rectangular cross section. Let the beam be subjected to deforming forces so that it bends (Fig. 6.13).

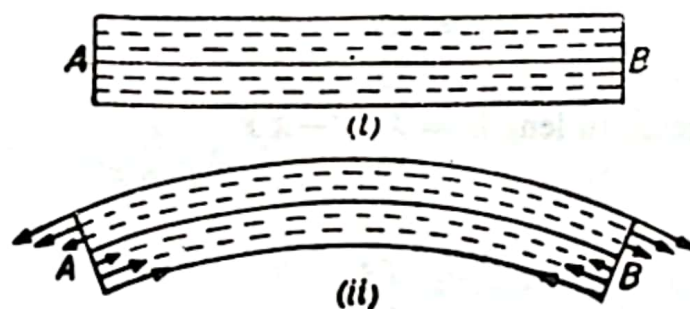


Fig. 6.13

In the initial position of the beam, the various filaments constituting the beam are in parallel layers of equal length. When the deforming forces are applied, the beam bends as shown in Fig. 6.13 (ii). Above the layer AB , the filaments are elongated while below AB they are compressed. The length of the layer AB remains unaltered. It is called the **neutral axis**. The surface containing the neutral axis and perpendicular to the plane of bending is called the **neutral surface**. Further the change in length of any filament (extension or contraction) is proportional to the distance of the filament from the neutral axis.

6.19 Bending Moment

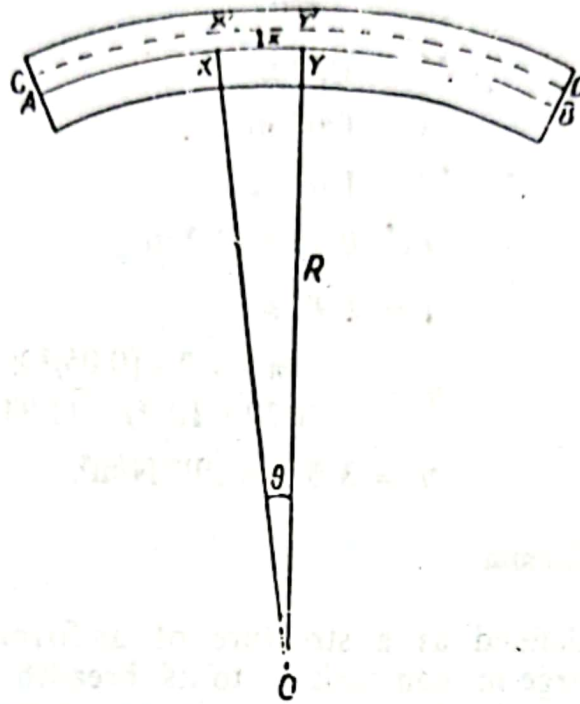


Fig. 6.14

Consider a beam under the action of deforming forces. Due to the property of elasticity of the material, a restoring couple acts on the beam. In the equilibrium position, the bending couple is equal and opposite to the restoring couple. *The moment of the restoring couple is called the bending moment.*

Consider a filament $X'Y'$ at a distance x from the neutral axis. Here $X'Y'$ has been extended. In Fig. 6.14,

$$XY = R\theta$$

and

$$X'Y' = (R+x)\theta$$

$$\therefore \text{Increase in length} = X'Y' - XY$$

$$= (R+x)\theta - R\theta = x\theta$$

Strain

$$= \frac{x\theta}{R\theta} = \frac{x}{R}$$

$$\text{Young's modulus } Y = \frac{\text{Stress}}{\text{Strain}}$$

$$\text{Stress} = Y \times \text{Strain} = \frac{Yx}{R}$$

The cross section of the beam is shown in Fig. 6.15.

The force acting on the element of area of cross section δa

$$= \text{stress} \times \delta a$$

$$F = \frac{Yx}{R} \times \delta a$$

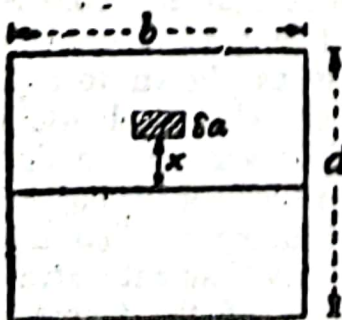


Fig. 6.15

The forces producing elongation act on the upper half outward and those producing contraction act on the lower half inward. These two forces constitute a couple.

Moment of the force about the neutral axis.

$$= \left[\frac{Yx}{R} \times \delta a \right] \cdot x = \left(\frac{Yx^2}{R} \right) \delta a$$

The moment of all the forces about the neutral axis

$$\begin{aligned} &= \sum \frac{Yx^2}{R} \delta a \\ &= \frac{Y}{R} \sum x^2 \delta a \end{aligned}$$

$$\text{Here } \sum x^2 \delta a = aK^2 = I_g$$

where a is the area of cross section of the beam and K is the radius of gyration and $aK^2 = I_g$. I_g is called the geometrical moment of inertia of the beam. Hence, Bending moment = $\frac{YI_g}{R}$

Special Cases

1. Rectangular cross section

If the breadth of the beam is b and thickness is d , then

$$a = b \times d$$

$$\text{and } K^2 = \frac{d^2}{12}$$

$$\therefore I_g = aK^2 = \frac{(bd)d^2}{12} = \frac{bd^3}{12}$$

2. Circular cross section

$$\text{Here } a = \pi r^2 \text{ and } K^2 = \frac{r^2}{4}$$

$$\therefore I_g = aK^2 = \frac{\pi r^2 \times r^2}{4} = \frac{\pi r^4}{4}$$

Flexural rigidity. It is defined as the external bending moment required to produce a unit radius of curvature. Therefore, flexural Rigidity = YI_g .

6.20 Basic Assumptions for Theory of Bending

1. The length of the beam is large in comparison to the thickness of the beam. Consequently, the shearing stresses are negligible.

2. The stress is proportional to strain. For elongation as well as compression, the Young's modulus of the beam has the same value.

3. The cross section of the beam remains unchanged during bending and hence the value of the geometrical moment of Inertia I_e is constant.

4. The applied forces are large in comparison to the weight of the beam.

5. Curvature of the beam. For a short length dS of the beam, the curvature at a point is given by

$$\frac{1}{R} = \frac{d\theta}{dS}$$

Also the slope $\theta = \tan \theta = \frac{dy}{dx}$

$$\therefore \frac{1}{R} = \frac{d}{dS} \left(\frac{dy}{dx} \right)$$

For a small element of the beam

$$dS = dx$$

$$\therefore \frac{1}{R} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$$

$$\frac{1}{R} = \frac{d^2y}{dx^2} = \text{Rate of change of slope.}$$

6.21 ✓ Cantilever

A cantilever is a thin uniform bar fixed horizontally at one end and loaded at the other end (Fig. 6.16). Let AB represent the

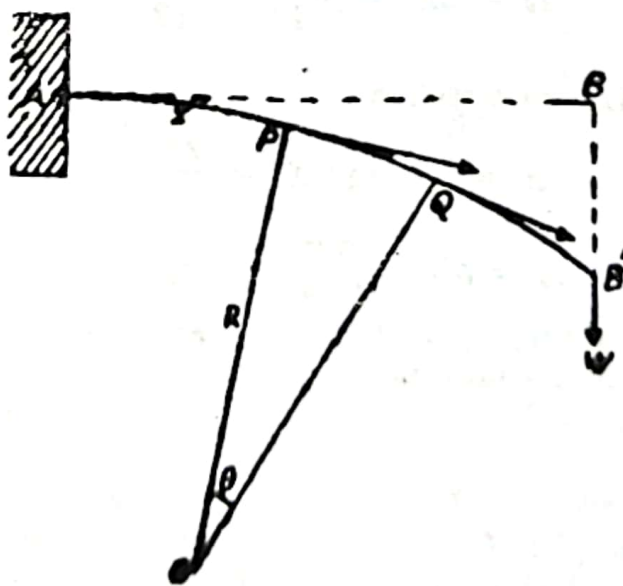


Fig. 6.16

neutral axis of the cantilever of length l . The end A is fixed and the end B is loaded with a load W vertically downwards. The end B is displaced to the position B' . The neutral axis of the cantilever

shifts to the position AB' . It is assumed that the weight of the cantilever is negligible.

Consider an element PQ , at a distance x from the end A and of radius of curvature R . Bending moment

$$= \frac{YI_\theta}{R}$$

Deflecting couple $= W(l-x)$

For equilibrium

$$\frac{YI_\theta}{R} = W(l-x) \quad \dots(i)$$

But $\frac{1}{R} = \frac{d^2y}{dx^2}$

$$\therefore \frac{d^2y}{dx^2} = \frac{W(l-x)}{YI_\theta} \quad \dots(ii)$$

Integrating, equation (ii)

$$\frac{dy}{dx} = \left(\frac{W}{YI_\theta}\right)\left(lx - \frac{x^2}{2}\right) + k_1$$

When $x = 0$, $\frac{dy}{dx} = 0$. $\therefore k_1 = 0$

$$\therefore \frac{dy}{dx} = \left(\frac{W}{YI_\theta}\right)\left(lx - \frac{x^2}{2}\right) \quad \dots(iii)$$

Integrating equation (iii)

$$y = \left(\frac{W}{YI_\theta}\right)\left(\frac{lx^2}{2} - \frac{x^3}{6}\right) + k_2$$

When $x = 0$, $y = 0$, $\therefore k_2 = 0$

$$\therefore y = \left(\frac{W}{YI_\theta}\right)\left(\frac{lx^2}{2} - \frac{x^3}{6}\right) \quad \dots(iv)$$

For the depression of the free end, $x = l$

$$\therefore y = \left(\frac{W}{YI_\theta}\right)\left(\frac{l^3}{2} - \frac{l^3}{6}\right)$$

$$y = \frac{Wl^3}{3YI_\theta} \quad \dots(v)$$

Special cases

1. Rectangular cross-section

$$I_\theta = \frac{bd^3}{12}$$

$$\therefore y = \frac{4Wl^3}{Ybd^3} \quad \dots (vi)$$

2. Circular cross:section

$$I_0 = \frac{\pi r^4}{4}$$

$$\therefore y = \frac{4Wl^3}{3Y\pi r^4} \quad \dots (vii)$$

$$Y = \frac{4Wl^3}{3\pi r^4 y} \quad \dots (viii)$$

Heavy Cantilever

In case the weight of the cantilever W_1 cannot be neglected the depression for the free end is given by

$$\frac{YI_0}{R} = W(l-x) + \left(\frac{W_1}{2l} \right) (l-x)^2$$

Here $\frac{W_1}{l} (l-x)$ is the weight of the beam from the point P upto the end B and $\frac{(l-x)}{2}$ is the distance of this weight from the end B' .

$$\text{But,} \quad \frac{1}{R} = \frac{d^2 y}{dx^2}$$

$$\therefore \frac{d^2 y}{dx^2} = \left(\frac{1}{YI_0} \right) \left[W(l-x) + \left(\frac{W_1}{2l} \right) (l-x)^2 \right] \dots (i)$$

Integrating twice, we get

$$y = \left(\frac{1}{YI_0} \right) \left[\frac{Wl^3}{3} + \frac{W_1 l^3}{8} \right] \quad \dots (ii)$$

$$y = \left(\frac{l^3}{3YI_0} \right) \left[W + \left(\frac{3W_1}{8} \right) \right] \quad \dots (iii)$$

$$\therefore Y = \left(\frac{l^3}{3YI_0} \right) \left[W + \left(\frac{3W_1}{8} \right) \right] \quad \dots (iv)$$

Note. If the cantilever bends by its own weight, $W = 0$ and

$$y = \frac{W_1 l^3}{8YI_0} \quad \dots (v)$$

6.22 Beam supported at its Ends and Loaded in the Middle

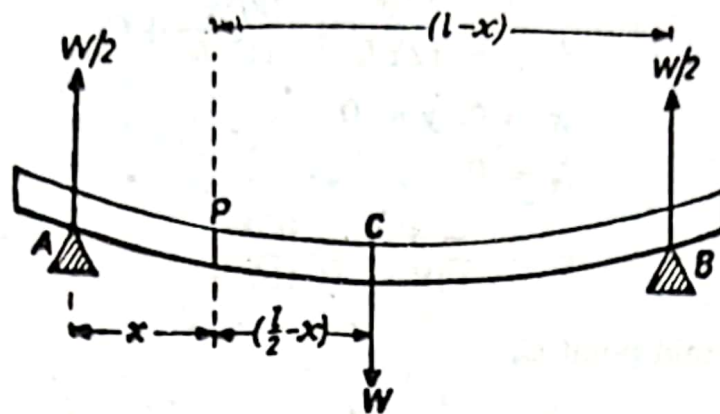


Fig. 6.17

Consider a beam (of negligible mass) supported on the two knife edges A and B (Fig. 6.17). The length of the beam between the two points A and B is l . It is loaded in the middle with a load W . The reaction at each support is $W/2$ in the vertically upward direction. Consider a cross section of the beam at P , at a distance x from A . For equilibrium

$$\frac{YI_0}{R} = W \left[\frac{l}{2} - x \right] - \frac{W}{2} (l - x)$$

$$\frac{YI_0}{R} = -\frac{W}{2} \times x$$

$$\therefore \frac{1}{R} = -\frac{Wx}{2YI_0} \quad \dots(i)$$

But $\frac{1}{R} = \frac{d^2y}{dx^2}$

$$\therefore \frac{d^2y}{dx^2} = -\frac{Wx}{2YI_0} \quad \dots(ii)$$

Integrating with respect to x

$$\frac{dy}{dx} = -\frac{Wx^2}{4YI_0} + C_1 \quad \dots(iii)$$

At $x = \frac{l}{2}$, $\frac{dy}{dx} = 0$

$$0 = -\frac{Wl^2}{16YI_0} + C_1$$

$$C_1 = \frac{Wl^2}{16YI_0} \quad \dots(iv)$$

Substituting this value in equation (iii)

$$\frac{dy}{dx} = -\frac{Wx^2}{4YI_0} + \frac{Wl^2}{16YI_0}$$

Integrating with respect to x

$$y = -\frac{Wx^3}{12YI_0} + \frac{Wl^2x}{16YI_0} + C_1 \quad \dots(v)$$

At $x = 0, y = 0$

$$\therefore C_1 = 0$$

$$\therefore y = -\frac{Wx^3}{12YI_0} + \frac{Wl^2x}{16YI_0} \quad \dots(vi)$$

At the mid point C ,

$$\frac{x}{2} = \frac{l}{2} \text{ and}$$

$$y = -\frac{Wl^3}{96YI_0} + \frac{Wl^3}{32YI_0}$$

$$y = \frac{Wl^3}{48YI_0} \quad \dots(vii)$$

(1) For a beam of rectangular cross-section

$$I_0 = \frac{bd^3}{12}$$

$$\therefore y = \frac{Wl^3}{4Ybd^3} \quad \dots(viii)$$

(2) For a beam of circular cross section

$$I_0 = \frac{\pi r^4}{4}$$

$$\therefore y = \frac{Wl^3}{12\pi Yr^4} \quad \dots(ix)$$

6.23 Depression of Beam under its Own Weight

Let a heavy beam of mass w per unit length be supported at the points A and B . The weight of the beam between the two knife edges, $W_1 = wl$.

The reaction at each support $= \frac{wl}{2}$ in the vertically upward direction. The beam bends under its own weight (Fig. 6.18).

Consider a cross-section of the beam at P at a distance x from A . The weight of the portion PB of the beam is $w(l-x)$. This weight acts at $w\left(\frac{l-x}{2}\right)$ from P . The reaction $\frac{wl}{2}$ at B produces a deflection moment.

The value of $I = \text{Mass} \cdot \left(\frac{L^2 + B^2}{12} \right)$

$$I = m \left(\frac{L^2 + B^2}{12} \right)$$

Here m is the mass, L the length and B the breadth of each bar.

(iii) Knowing the values of Y and η , the values of bulk modulus of elasticity can be calculated from the equation.

$$K = \frac{Y\eta}{9\eta - 3Y} \quad \dots(v)$$

(iv) Poisson's ratio,

$$\sigma = \left(\frac{Y}{2\eta} - 1 \right) \quad \dots(vi)$$

Substituting the values of Y and η , the value of σ is given by

$$\sigma = \frac{T_2^2}{2T_1^2} - 1$$

$$\sigma = \left(\frac{T_2^2 - 2T_1^2}{2T_1^2} \right) \quad \dots(vii)$$

The advantages of this method are :

(i) the values of Y and η can be determined by measuring the time period.

(ii) Only a small length of the specimen in the form of a wire is required.

(iii) All the four constants can be determined accurately because the time periods can be measured accurately.

Exercises 6

1. Define the terms yield point, elastic limit, elastic fatigue and breaking stress.

2. Define Poisson's ratio. Describe an experiment with necessary theory to determine the Poisson's ratio for rubber.

[Delhi (Sub) 1976]

3. Show that the work done per unit volume in straining a body

is equal to $\frac{1}{2}$ (stress $\times$ strain) [Delhi (Sub) 1980]

4. Explain the terms elastic limit, rigidity modulus and Poisson's ratio. Show that in the case of longitudinal strain, the work done per unit-volume is half the product of stress and strain.

[Calcutta 1980]

5. Prove that a shear is equivalent to two linear strains of half the magnitude at right angles to each other.

[Calcutta (Hons) 1980]

6. Show that the bulk modulus of elasticity

$$K = \frac{Y}{3(1 - 2\sigma)} \quad [\text{Delhi (Sub) 1980}]$$

7. Derive the relation

$$\eta = \frac{Y}{2(1+\sigma)} \quad [\text{Delhi (Sub) 1980}]$$

8. Show that

$$\sigma = \left(\frac{Y}{2\eta} \right) - 1$$

9. Prove that

$$\frac{3}{Y} = \frac{1}{3K} + \frac{1}{\eta}$$

10. Explain the terms Poisson's ratio and discuss its limiting values. [Delhi (Sub) 1980]

11. Define young's modulus, modulus of rigidity and Poisson's ratio. [Calcutta, 1978]

12. Describe a method for the determination of the modulus of rigidity of the material of a solid cylinder. Deduce the formulae you would use. How will the expression for the moment of couple acting on the cylinder be modified in the case of a hollow cylinder. [Calcutta, 1979]

13. Derive an expression for the torsional rigidity of a wire.

14. Show that the couple required per unit twist in the case of a cylinder is

$$C = \frac{\pi \eta r^4}{2l} \quad [\text{Mysore 1977}]$$

15. Describe briefly the statical method for determining the modulus of rigidity of a wire.

16. Give the theory and method for determining the modulus of rigidity of a wire using a torsion pendulum.

17. Give the Maxwell's needle method for determining the modulus of rigidity of a wire.

18. Show that the bending moment for a thin uniform bar of rectangular cross-section is

$$\frac{Ybd^3}{12R}$$

19. What do you understand by the term geometrical moment of inertia. Give its values for (i) rectangular cross-section and (ii) circular cross-section.

20. Derive an expression for the depression of the free end of a cantilever fixed at one end and loaded at the other end.

21. Give the theory of bending and show that the depression at the free end of a cantilever is given by

$$y = \left(\frac{l^3}{3YI_0} \right) \left[W + \left(\frac{3}{8} W_1 \right) \right]$$

Here W_1 is the weight of the cantilever.

22. A thin uniform bar of rectangular cross-section is supported at its ends on two knife edges and loaded in the middle. Derive an expression for the depression of the mid-point of the beam for a load W . Neglect the mass of the beam.

[Delhi (Sub) 1976, Mysore 1978]

23. Derive an expression for the depression of the mid point of a beam of weight W_1 supported at the ends and loaded in the middle with a load W .

24. What is an I-section girder? Why are I-section girders preferred?

25. Describe Koenig's method for determining Young's modulus of the material of a beam.

26. A thin uniform bar is clamped at one end and loaded at the other. The free end is displaced and released. Calculate the time period of vibration of the free end.

27. Give briefly the method of determining the Young's modulus of the material of a beam by vibration method. Take into account the mass of the beam.

28. Give the method of determining the elastic constants Y , η and σ by Searle's method.

29. Define Young's modulus, bulk modulus of rigidity and Poisson's ratio. (Delhi, 1976)

30. Show that

$$Y = \frac{9K\eta}{3K + \eta}$$

(Delhi 1976)

31. Show that

$$\eta = \frac{Y}{2(1 + \sigma)}$$

[Delhi (Sub) 1976, 77]

32. Define the elastic constants. Establish relations between them. (Delhi (Hons) 1976)

33. What are torsional oscillations? Derive an expression for the time period of such oscillations. (Delhi (Sub) 1977)

34. Calculate the strain in a rod of isotropic material stretched in such a way that all lateral strains are prevented.

(Delhi (Hons) 1977)

[Hint. Here $\beta = \frac{\alpha}{2}$ and $\sigma = 0.5$, It is an ideal case.]

35. Derive an expression for the restoring couple acting on a cylinder of length l and radius a when its top end is clamped and the lower end is twisted through an angle θ about a vertical axis. Describe an accurate method for the determination of modulus of rigidity of the material of the wire.

(Delhi (Hons) 1977)

36. Show that

$$\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{K}$$

(Delhi 1978)

37. Define a cantilever. Obtain an expression for the depression produced at its free end when the weight of the beam is negligible. [Delhi, 1978] [Delhi (Sub) 1978]

38. Show that shear is equivalent to an elongation strain and a compression strain. (Delhi 1979)

39. Show that for a homogeneous isotropic medium

$$Y = 2\eta(1 + \sigma) \quad (\text{Delhi 1979})$$

40. Derive an expression for the couple required to bend a uniform straight metallic strip into an arc of a circle of small curvature. (Delhi 1979)

41. Explain what is meant by modulus of rigidity of a material. Give the details of the Maxwell's needle method for the determination of the modulus of rigidity for the material of a wire. [Delhi (Sub) 1979]

42. Show that

$$Y = \frac{9\eta K}{3K + \eta} \quad [\text{Delhi (Hons) 1979}]$$

43. Derive an expression for the Bending Moment of a horizontal beam clamped at one end and loaded at the other and hence derive an expression for the depression of a beam supported at the ends and loaded at the centre. neglecting the weight of the beam. [Delhi, 1982]

44. (a) Define the terms yield point, elastic limit, elastic fatigue and breaking stress.

(b) Show that modulus of rigidity η , Young's modulus Y and Poisson's ratio σ are related by the equation

$$\eta = \frac{Y}{2(1 + \sigma)} \quad [\text{Delhi (Sub) 1982}]$$

45. Deduce the relation between elastic constants

$$Y, k, \eta \text{ and } \sigma \quad [\text{Meerut, 1982}]$$

46. Define shearing stress and shearing strain. Calculate the torque required to twist a cylinder of length l and radius r through an angle θ . [Lucknow, 1982]

47. A metallic beam is supported at both ends and loaded in the middle. Calculate the depression of the middle point. [Rohilkhand, 1982]

48. A beam of rectangular cross section is fixed with one end in a vertical wall. Obtain an expression for the depression at the other end which is loaded. [Delhi, 1982]